IBM FileNet P8 V5.5.8

Detailed Study Material

Day 7 – Workflow Development using Process Designer

Process Designer | Workflow Design | Components | Build & Deploy

Author	Course Day	Version
Rajesh Pasham	**Day 7 of 10**	**IBM FileNet P8 V5.5.8**

Section 1 | Learning Objectives

By the end of Day 7, students will be able to:

1	Describe IBM Process Designer and its role in creating workflow definitions for FileNet P8 V5.5.8
2	Identify and explain the four core workflow design components: Process Maps, Workflow Steps, Routing Logic, and Conditional Branches
3	Design a complete Process Map with Launch Step, Queue Steps, System Steps, Routing Decisions, and Terminate step
4	Configure Workflow Components: Activities, Sub-processes, Escalations, and Timers
5	Define workflow data fields (F_Subject, custom fields) and configure their data types and default values
6	Implement conditional routing using PE Boolean expressions to branch document approval flows
7	Configure Escalation rules and Timer deadlines to handle overdue work items automatically
8	Build a complete 4-step Document Approval Workflow: Submit, Review, Approve/Reject, Notify
9	Export the workflow map as a .pep file and deploy it to the FileNet P8 V5.5.8 Process Engine
10	Verify the deployed workflow by launching a test instance and checking queue activity in ACCE

Section 2 | Process Designer Introduction

2 Process Designer Introduction

IBM Process Designer is a standalone Java-based desktop application used to create, edit, and deploy workflow definitions for IBM FileNet P8 V5.5.8. It provides a visual drag-and-drop canvas for building workflow maps — the templates that define how business processes route documents through approval, review, and processing steps.

Process Designer Key Facts

- Installed separately from CE and PE: IBM Installation Manager > Process Designer package.
- Runs as a fat-client GUI application on Windows (ProcessDesigner.exe).
- Connects directly to the FileNet P8 Process Engine (PE) on port 32768.
- Workflow maps are saved as .pep files (Process Engine Program files).
- Deployment: Process Designer deploys the .pep file directly to PE via the PE connection.
- No coding required for standard workflows. Advanced steps use JavaScript or Java.
- Training lab connection: Process Designer connects to localhost:32768 (PE_ConnectionPoint).

2.1 Process Designer Interface

Interface Element	Description
Process Map Canvas	The main design area where workflow steps and connectors are placed. Steps are added by dragging from the palette to the canvas.
Step Palette	Left panel listing all available step types: Launch, Queue, System, Map, Notify, Script, Abort, Milestone, Component.
Properties Panel	Right panel showing properties for the selected step or connector. Fields, routing expressions, deadline settings.
Data Fields Panel	Bottom panel listing all workflow data fields defined for this workflow. Shows name, data type, and default value.
Connector Tool	Arrow tool for drawing connections between steps. Connectors carry routing expressions for conditional branching.
Toolbar	Save, Open, Deploy to PE, Validate, Print, and Zoom controls.
Map Hierarchy	Tree view showing the current workflow map and any sub-maps (sub-processes) it contains.

2.2 Process Designer Installation

Installing IBM Process Designer V5.5.8

1. Run IBM Installation Manager on the developer workstation (Windows).
2. Add the Process Designer repository: ProcessDesigner_5.5.8_WIN.zip > repository.config
3. Install to: C:\IBM\FileNet\ProcessDesigner\
4. Launch: C:\IBM\FileNet\ProcessDesigner\ProcessDesigner.exe
5. On first launch: File > Connection > New Connection
   - Connection Name: TrainingPEConnection
   - Server: localhost

- Port: 32768
- User: ceadmin
- Password: ceadmin

6. Click Test Connection to verify PE connectivity before designing.

Section 3 | Workflow Design Components

3 Workflow Design Components

Every FileNet P8 workflow is built from four fundamental design components: Process Maps, Workflow Steps, Routing Logic, and Conditional Branches. Understanding each component is essential before building any workflow.

3.1 Process Maps

A Process Map is the complete visual definition of a workflow. It is the canvas in Process Designer where all steps, connectors, data fields, and routing rules are assembled. Each Process Map produces one .pep file when saved.

3.1.1 Process Map Properties

Property	Description
Map Name	Unique identifier for the workflow. Example: DocumentApprovalWorkflow. No spaces. Used by Event Subscriptions and API calls to launch the workflow.
Map Description	Human-readable description shown in ICN and ACCE workflow monitors.
Launch Step	The entry point. Every process map must have exactly one Launch Step. Defines required data fields for workflow launch.
Data Fields	Custom fields defined at the map level: name, data type (String, Integer, Date, Boolean, Float), initial value, and optionally hidden/visible.
Version	Every time a modified map is deployed, PE increments the version. Active instances continue on their current version.
PE Connection	Which PE server this map is deployed to. In training: localhost:32768 (PE_ConnectionPoint).

3.1.2 Workflow Data Fields

Data Fields are variables that travel with a workflow instance (Work Object). They are defined in the Process Map and accessible at every step.

Standard System Data Fields (Auto-created by PE)

F_Subject (String) — Primary subject/title of the workflow. REQUIRED at launch.
F_BoundDocument (String) — GUID of the attached FileNet document. Set by Process Router.
F_Originator (String) — Username of who launched the workflow.
F_Deadline (DateTime)— Optional deadline for the entire workflow instance.
F_StepName (String) — Read-only: the name of the current step.
F_Comment (String) — Optional free-text comment from the current user.
F_Overdue (Boolean) — Set to true by PE when F_Deadline has passed.

In addition to system fields, you define custom fields for your business logic:

Field Name	Data Type	Purpose
F_ApprovalStatus	String	Approved / Rejected / Pending — tracks approval decision
F_ReviewerComments	String	Free-text comments from the reviewer step
F_Priority	String	High / Medium / Low — controls routing to fast-track queue
F_DocumentType	String	Invoice / Contract / HR — routes to specialist queues
F_ApprovedBy	String	Username of the approver who approved the document
F_RejectionCount	Integer	Number of times document has been rejected (loop control)
F_ApprovalDate	DateTime	Date/time when final approval was granted
F_RequiresLegalReview	Boolean	Flag for routing to Legal Review queue

3.2 Workflow Steps

Workflow Steps are the individual actions in a process map. Each step type has specific properties and behaviours. Steps are connected by Connectors (arrows) to form the workflow flow.

3.2.1 Step Types and Icons

Step Type	Description	Use Case
Launch Step	Entry point. Rectangular with green play icon. Exactly one per map. Defines fields required at launch.	E3F2FD
Queue Step	Rectangular with inbox icon. Assigns work item to a named queue for human action. Most common step.	FFFFFF
System Step	Rectangular with gear icon. Executes automatically. Contains scripts, component steps, or map steps.	FFFFFF
Routing Decision	Diamond shape. Evaluates a Boolean expression. Routes to different branches based on true/false result.	E3F2FD
Notify Step	Rectangular with envelope icon. Sends email via PE Mailer. No queue assignment.	FFFFFF
Map Step (Sub-map)	Rectangular with map icon. Launches a sub-workflow. Can	FFFFFF

	wait for completion before continuing.	
Milestone Step	Rectangular with flag icon. Sets a named progress milestone on the workflow instance.	E3F2FD
Abort Step	Rectangular with stop icon. Terminates the workflow immediately. Status: TERMINATED.	FFFFFF
Terminate Step	Circle. End of workflow. Sets status to COMPLETED. Every branch must eventually reach Terminate.	FFFFFF

3.2.2 Queue Step Configuration

The Queue Step is the most important step in most business workflows. It places a work item in a queue for a human user to action.

Property	Description
Queue Name	The target queue. Example: TrainingApprovalQueue. Must exist in the TrainingRoster on the connected PE.
Step Label	Human-readable name shown in ICN work list. Example: 'Manager Approval Required'.
Response Field	The data field that captures the user's decision. Usually F_ApprovalStatus with values Approved/Rejected.
Deadline Expression	PE expression to set the step deadline. Example: Dates.add(Dates.currentDate(), 2, 'd') = 2 days from now.
Escalation	Action to take when deadline passes: reassign to a different user, move to escalation queue, or send email.
Instructions	Optional text displayed to the user in ICN explaining what action to take.
Required Fields	Fields the user MUST fill before stepping the item. Process Designer enforces this validation.

3.3 Routing Logic

Routing Logic defines how work items travel from one step to the next. In Process Designer, routing is expressed through Connectors between steps.

3.3.1 Connector Types

Connector Type	Behavior
Unconditional Connector	A simple arrow with no expression. Work items always follow this path. Used for mandatory sequential steps.
Conditional Connector	An arrow with a Boolean expression. Work items follow this path only when the expression evaluates to TRUE.

Default Connector	A dashed arrow. Followed when no other conditional connector on this step evaluates to TRUE. Each routing decision should have exactly one default connector.
Error Connector	Routes to an error handling step when a system step throws an exception. Configured in the step's exception properties.

3.3.2 Routing Expression Syntax Reference

```
// String field comparisons
F_ApprovalStatus == "Approved"
F_ApprovalStatus == "Rejected"

// Numeric comparisons
F_RejectionCount < 3           // Allow up to 3 rejections before auto-terminate
F_RejectionCount >= 3          // Terminate after 3 rejections

// Boolean fields
F_RequiresLegalReview == true
F_IsConfidential == false

// String functions
Strings.upper(F_Priority) == "HIGH"
Strings.contains(F_DocumentType, "Invoice")

// Null / empty checks
F_ApprovedBy != null && F_ApprovedBy != ""

// Date comparisons
Dates.diff(F_Deadline, Dates.currentDate()) < 0   // Overdue

// Logical operators
F_Priority == "High" && F_ApprovalStatus == "Pending"
F_DocumentType == "Invoice" || F_DocumentType == "PO"
```

3.4 Conditional Branches

Conditional Branches are created by adding a Routing Decision step (diamond) after a Queue or System step, then drawing conditional connectors from the diamond to different next steps.

3.4.1 Single-Level Branch Pattern

Single-Level Branch: Approve or Reject

```
LaunchStep
  |
  v (unconditional)
ReviewQueue  (ceadmin reviews document)
  |
  v (unconditional)
Routing Decision (diamond)
  |-- Condition: F_ApprovalStatus == "Approved"  --> PublishStep
  |-- Condition: F_ApprovalStatus == "Rejected"  --> RevisionQueue
  |-- Default (no other conditions true)         --> AdminNotifyStep
```

3.4.2 Multi-Level Branch Pattern (Approval + Priority)

Multi-Level Branch: Priority-Based Routing Before Approval

```
LaunchStep
  |
  v
Routing Decision 1 (Priority check)
  |-- F_Priority == "High"   --> HighPriorityApprovalQueue (2-hour SLA)
  |-- F_Priority == "Medium" --> StandardApprovalQueue (2-day SLA)
  |-- Default                --> StandardApprovalQueue
  |
  (both queues lead to...)
  v
Routing Decision 2 (Approval result)
  |-- F_ApprovalStatus == "Approved" --> PublishStep --> Terminate
  |-- F_ApprovalStatus == "Rejected" && F_RejectionCount < 3 --> RevisionQueue (loop)
  |-- F_ApprovalStatus == "Rejected" && F_RejectionCount >= 3 --> AutoRejectStep --> Terminate
```

Section 4 | Workflow Components

4 Workflow Components

4.1 Activities

In Process Designer terminology, an Activity is a logical grouping of steps within a workflow map that represents a distinct phase of the business process. Activities improve readability and support milestone tracking.

Document Approval Workflow – Activity Breakdown

Activity 1: SUBMISSION — LaunchStep, InitializeFieldsStep, SubmissionNotifyStep
Activity 2: REVIEW — ReviewerQueue, ReviewDecisionStep
Activity 3: APPROVAL — ManagerApprovalQueue, ApprovalDecisionStep
Activity 4: POST-APPROVAL — PublishStep OR RevisionQueue OR AutoRejectStep
Activity 5: NOTIFICATION — SubmitterNotifyStep (Approved/Rejected email)
Activity 6: COMPLETION — MilestoneStep, TerminateStep

Activity Property	Description
Activity Name	Human-readable label. Shown in ACCE Workflow Monitor as the current phase.
Steps in Activity	Steps logically grouped in this activity. Each step belongs to one activity.
Milestone	Each activity can have a named milestone that is set when the activity begins. Used for SLA tracking.
Colour Coding	In Process Designer, different activities can use different background colours for visual clarity.

4.2 Sub-processes (Map Steps)

A Sub-process (Map Step) allows one workflow to launch another workflow as a child process. The parent workflow can either wait for the child to complete (synchronous) or continue independently (asynchronous).

4.2.1 Sub-process Configuration

Property	Description
Sub-map Name	The name of the child workflow definition to launch. Must be deployed on the same PE server.
Synchronous	Parent workflow waits until the child workflow reaches its Terminate step before proceeding.
Asynchronous	Parent workflow continues immediately after launching the child. Child runs independently.
Data Field Mapping	How parent workflow data fields are passed to the child workflow on launch.

Return Fields	Data fields the child workflow passes back to the parent on completion (synchronous only).

4.2.2 When to Use Sub-processes

- Legal Review: When F_RequiresLegalReview = true, launch a LegalReviewWorkflow as a synchronous sub-process. Parent waits for legal clearance.
- Parallel Signatures: Launch multiple SignatureWorkflows in parallel, then join when all are complete.
- Reusable Logic: Extract common business logic (e.g., notification pattern) into a sub-process reused by multiple parent workflows.
- Complex Approval Chains: Department head approval, then director approval — each level as a separate sub-process.

4.3 Escalations

Escalations are automatic actions triggered when a work item in a queue has exceeded its deadline. They prevent work items from being forgotten when users are unavailable.

4.3.1 Escalation Types

Escalation Type	Description	Example
Reassignment	Automatically reassign the work item to a different user or group when deadline expires.	Example: Reassign to supervisor if manager doesn't act within 2 days.
Queue Move	Move the work item to a different (escalation) queue where a supervisor can see all overdue items.	Example: Move to ExecutiveApprovalQueue after 48 hours.
Email Notification	Send an email alert to the assigned user and/or their supervisor when deadline is approaching or passed.	Example: 'Reminder: Document approval overdue' email to manager.
Abort (Terminate)	Terminate the workflow instance automatically if no action is taken within the escalation window.	Used for automated expiry of time-sensitive requests.

4.3.2 Escalation Configuration in Process Designer

1. Select the Queue Step that needs escalation
2. In the Properties panel, click the Deadline / Escalation tab
3. Set Deadline Expression (PE expression calculating when the step is due):

```
Dates.add(Dates.currentDate(), 2, 'd')    // 2 days from step entry
```

4. Set Escalation Action: Reassign, Move Queue, Send Email, or Abort
5. For Reassignment: enter the target user/group in the Reassign To field
6. Set Escalation Delay (how long after deadline before escalation fires): Example: 4 hours
7. Optionally set a Pre-deadline Reminder: send email 1 day before deadline

4.4 Timers

Timers introduce time-based waiting into workflow steps without requiring user action. A Timer causes a System Step to pause execution for a specified duration before proceeding.

4.4.1 Timer Use Cases

- Waiting Period: After document submission, wait 24 hours before auto-forwarding to approver (cooling-off period).
- Retry Logic: If an external system call fails, wait 30 minutes and retry automatically.
- SLA Countdown: Set a System Step as a timer that waits until business day deadlines.
- Batch Processing: Group documents submitted within a 4-hour window before sending for bulk approval.

4.4.2 Timer Configuration

```
// Timer expressions (PE expression language)

// Wait 24 hours from now
Dates.add(Dates.currentDate(), 24, 'h')

// Wait until next Monday 9:00 AM
Dates.nextDay(Dates.currentDate(), 2, 9, 0)   // 2 = Monday

// Wait until F_Deadline - 1 day (1 day before the workflow deadline)
Dates.add(F_Deadline, -1, 'd')

// Calculate 2 business days (skipping weekends)
Dates.addBusinessDays(Dates.currentDate(), 2)
```

Section 5 | Design Patterns & Best Practices

5 Workflow Design Patterns & Best Practices

5.1 Document Approval Workflow Pattern

The Document Approval Workflow is the most common FileNet P8 workflow pattern. Every organisation uses some variation of it. The training lab builds this exact pattern.

Training Workflow Map: DocumentApprovalWorkflow

```
DATA FIELDS:
  F_Subject          (String)  — Document title (system field, required at launch)
  F_BoundDocument    (String)  — FileNet document GUID (set by Process Router)
  F_ReviewerComments (String)  — Comments from reviewer
  F_ApprovalStatus   (String)  — Approved / Rejected / Pending (default: Pending)
  F_ApprovedBy       (String)  — Username of approver
  F_RejectionCount   (Integer) — Rejection counter (default: 0)
  F_Priority         (String)  — High / Medium / Low (default: Medium)

STEPS (in order):
  1. LaunchStep            — Entry point, accepts F_Subject and F_Priority
  2. InitializeStep        — System Step: sets F_ApprovalStatus = 'Pending'
  3. SubmissionNotifyStep  — Notify: email submitter confirming receipt
  4. TrainingReviewQueue   — Queue Step: reviewer checks document quality
  5. ReviewDecision        — Routing Decision on F_ReviewerComments
  6. TrainingApprovalQueue — Queue Step: manager approves/rejects
  7. ApprovalDecision      — Routing Decision on F_ApprovalStatus
  8a. ApprovalNotifyStep   — Notify (if Approved): email submitter
  8b. RejectionDecision    — Check F_RejectionCount < 3
  9a. RevisionQueue        — Loop back (if Rejected, count < 3)
  9b. AutoRejectStep       — System Step: set status = AutoRejected (if count >= 3)
  10. MilestoneStep        — Sets milestone: 'ProcessComplete'
  11. TerminateStep        — End of workflow
```

5.2 Best Practices

Best Practice	Rationale
Always start with data fields	Define ALL data fields before placing steps. Adding fields later can break existing steps.
Name steps descriptively	Use names like 'ManagerApprovalRequired' not 'Step3'. Users see step names in ICN.
Every branch must reach Terminate	Ensure every conditional path eventually leads to a Terminate step. Orphaned steps cause runtime errors.
Use Default connectors	Every Routing Decision must have a Default connector for cases where no condition is true.

Validate before deploying	Use Process Designer's Validate Map function to catch errors before deploying to PE.
Version control .pep files	Save .pep files to version control (Git). Name with version: DocApproval_v1.2.pep.
Test with ceadmin account	Test the workflow with ceadmin credentials in ICN before releasing to end users.
Set reasonable deadlines	Use PE date expressions for deadlines. Hard-coded dates are fragile.

5.3 Deployment Process

After designing and validating the workflow map, it must be deployed to the Process Engine server so that workflows can be launched.

8. In Process Designer, click File > Save to save the .pep file locally
9. Click Map > Validate Map to check for design errors (no orphaned steps, valid expressions, etc.)
10. Fix any validation errors reported. Common errors: missing Terminate, invalid expression syntax, undefined data field
11. Click Map > Deploy to Process Engine
12. In the Deploy dialog, verify:

```
Connection: TrainingPEConnection (localhost:32768)
Map Name: DocumentApprovalWorkflow
Deploy as new version: Yes (for updates) / No (first deployment)
```

13. Click Deploy. Process Designer sends the .pep to PE and registers it
14. In ACCE > Process Configuration Console > Workflow Definitions, verify DocumentApprovalWorkflow is listed
15. In ACCE > TrainingOS > Events, Actions, Processes > Subscriptions, create a Process Router subscription to launch this workflow when a document is checked in

Section 6 | Troubleshooting & Quick Reference

6 Troubleshooting & Quick Reference

6.1 Common Process Designer Issues

Symptom	Cause	Resolution
Cannot connect to PE	Wrong server/port or PE service stopped	Verify PE service running. Check port 32768. Verify ceadmin credentials.
Validation error: orphaned step	A step has no incoming or outgoing connector	Connect the step to the workflow flow. Every step needs at least one in and one out.
Validation error: no Terminate	A branch path does not end at a Terminate step	Trace each conditional branch to confirm it reaches a Terminate step.
Deploy fails: map already exists	Deploying a map with the same name when version conflicts exist	Use 'Deploy as new version' option or delete the old definition from ACCE first.
Workflow not in ACCE after deploy	Deployment to wrong PE connection	Verify the PE connection in Process Designer points to the correct PE server.
F_BoundDocument not set	Workflow launched manually, not via Process Router	Set F_BoundDocument manually in the Launch Step or use a Process Router subscription.
Queue step not routing correctly	Expression evaluates to empty — default connector fires unexpectedly	Check data field default value. Add a System Step before routing to set field defaults.
Sub-process not launching	Child map not deployed to PE or name mismatch	Deploy the child map first. Verify map name matches exactly (case-sensitive).

6.2 Quick Reference

Item	Value	Notes
Process Designer exe	C:\IBM\FileNet\ProcessDesigner\ProcessDesigner.exe	Launch Process Designer client
Workflow file type	.pep (Process Engine Program)	Saved workflow map file
PE connection port	32768	RMI/IIOP port for Process Designer
Validate shortcut	Map > Validate Map	Check design before deploying
Deploy shortcut	Map > Deploy to Process Engine	Push .pep to PE server

View deployed workflows	ACCE > Process Config Console > Workflow Definitions	Verify deployment
Test launch workflow	ACCE > TrainingOS > Browse > doc > right-click > Launch Workflow	Manual test
Monitor active instances	ACCE > Process Config Console > Active Workflows	Live monitoring
PE log location	C:\IBM\FileNet\ProcessEngine\log\vwtaskd.log	Workflow execution errors
WAS log	C:\IBM\WAS\profiles\FileNetServer\logs\server1\SystemOut.log	JDBC and PE errors
Launch via REST API	POST http://localhost:9080/pe-rest/v1/workobjects	Programmatic launch